

DAWEI BAI

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Updated 12/2025

ACADEMIC POSITIONS

Postdoctoral Associate , Yale University, New Haven, CT, USA. Sponsors: Brian Scholl and Samuel McDougle	2023 – present
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EDUCATION

Ph.D. , École Normale Supérieure, Institut Jean Nicod, Paris, France. Supervisor: Brent Strickland	2019 – 2023
M.Sc., Cognitive Science (“Cogmaster”), École Normale Supérieure, École des Hautes Études en Sciences Sociales and Paris Descartes University, Paris, France.	2017 – 2019
M.A., Linguistics , University of Lille, Lille, France.	2016 – 2017
B.S., Psychology , University of Lille and University of Poitiers, Lille, France.	2014 – 2017
B.A., French Literature , University of Poitiers, Poitiers, France.	2013 – 2016
Engineering study , University of Technology of Compiègne, Compiègne, France.	2011 – 2013

PUBLICATIONS

- Bai, D.**, Hafri, A., Izard, V., Firestone, C., & Strickland, B. (2025). “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving [target article]. *Behavioral and Brain Sciences*. <https://doi.org/10.1017/S0140525X25102756>
- Bai, D.**, & Strickland, B. (2025). The “Double Ring Illusion”: The physical constraint of solidity shapes visual processing. *Journal of Experimental Psychology: General*, 154(12), 3417–3427. <https://doi.org/10.1037/xge0001839>
- Bai, D.**, & Strickland, B. (2023). The Pulfrich solidity illusion: A surprising demonstration of the visual system’s tolerance of solidity violations. *Psychonomic Bulletin & Review*, 30(5), 1782–1787. <https://doi.org/10.3758/s13423-023-02271-9>

PUBLICATIONS UNDER REVIEW

Bai, D., & Scholl, B. J. (under review). The tiger moving through the tall grass: Spontaneous perception and tracking of featureless objects.

WORKING PAPERS

Bai, D., Weisman, K., Schille-Hudson, E., Ihm, E., Taves, A., Luhrmann, T. M., & Scholl, B. J. (in preparation). Chasing the supernatural: The perception of animacy from motion is related to spiritual experiences.

Bai, D., & Scholl, B. J., & Strickland, B. (in preparation). Isolating event categories in space and time: Evidence of dynamic “on the fly” representations in perception.

Jones, H., **Bai, D.,** Scholl, B. J., & Awh, E. (in preparation). Electroencephalogram decoding of the attentional selection and tracking of featureless objects.

Sun, Z., **Bai, D.,** Scholl, B. J., & McDougle, S. (in preparation). Visuomotor adaptation via featureless objects.

INVITED TALKS

2025 “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving.
Chaz Firestone lab, Johns Hopkins University.

2025 The tiger moving through the tall grass: Spontaneous perception and tracking of featureless objects. Martin Rolfs lab, Humboldt-Universität zu Berlin.

2023 Intuitive physics in visual perception. Hans Op de Beeck lab, KU Leuven.

2023 From object perception to object cognition: Philosophical and empirical perspectives.
Memory and Mental Files workshop, Grenoble, France.

2021 The Solidity Principle in Visual Perception.
Language, Thought, and Perception Workshop, Nantes, France.

CONFERENCE PRESENTATIONS

Bai, D., Hafri, A., Izard, V., Firestone, C., & Strickland, B. (2026). “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving. Talk to be given at the *Budapest CEU Conference on Cognitive Development*, Budapest, Hungary.

Bai, D., Weisman, K., Schille-Hudson, E., Ihm, E., Taves, A., Luhrmann, T. M., & Scholl, B. (2025). Chasing the supernatural: The perception of animacy from motion is related to spiritual experiences. Poster presented at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, FL.

- Bai, D., & Scholl, B., (2024).** Object tracking without objects: Perceiving persistence defined by pure change. Poster presented at the *European Conference on Visual Perception*, Aberdeen, Scotland.
- Bai, D., & Scholl, B., (2024).** Do visual objects lose their individuality due to the perception of collective goals?: Evidence from numerical underestimation in the Wolfpack effect. Poster presented at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, FL.
- Bai, D., & Strickland, B. (2023).** Object solidity disambiguates ambiguous motion. Talk given at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, FL.
- Bai, D., & Strickland, B. (2022).** The double ring illusion: Object solidity is used to disambiguate ambiguous motion. Talk and demo night presentation given at the *European Conference on Visual Perception*, Nijmegen, Netherlands.
- Bai, D., & Strickland, B. (2022).** The double ring illusion: Object solidity is used to disambiguate ambiguous motion. Poster presented and demo night presentation at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, FL.
- Bai, D., Falck, A., & Strickland, B. (2021).** The adult visual system resists learning continuity violations. Poster presented at the *Virtual Vision Science Society Meeting*. Online.
- Bai, D., O'Madagain, C., Peperkamp, S., & Strickland, B. (2019).** Core Cognition and Learnability: A Possible Explanation for Cross-linguistic Similarities. Talk given at the *21st conference of the European Society for Cognitive Psychology*, Tenerife, Spain.

AWARDS

Third place in the “Best Illusion of the Year Contest” in 2021 by the Neural Correlate Society, with the “Double Ring Illusion”. View the illusion at <https://youtu.be/bVbPmFpspXc>.